Problem #1: The ρ - T plane (22 points)

Problem #1a: Setting boundaries (10 points)

Consider a gas of ionized hydrogen. Using the relations we derived in class find the boundary in the ρ - T plane where:

(i) radiation pressure dominates the equation of state of the gas (5 points)

(ii) the ideal gas pressure dominates the equation of state of the gas (5 points)

Problem #1b: Big stars and little stars (12 points)

(i) Using the mass - radius relation calculate the average density of a $10\,\mathrm{M}_\odot$ and $150\,\mathrm{M}_\odot$ star on the main sequence. (4 points)

(ii) Using the virial theorem as described in chapter two to calculate the average temperature of the $10\,\mathrm{M}_\odot$ and $150\,\mathrm{M}_\odot$ stars. (4 points)

(iii) Does radiation pressure or ideal gass pressure dominate for each star based on their average densities and temperatures? (4 points)

Problem #2: Writing an abstract from preliminary work (20 points)

Working as a researcher, you'll encounter situations where you need to write an abstract for a presentation on a research project that isn't finished yet. This can be kind of a scary thing to do if you aren't quite sure what you might actually end up finding; but... deadlines are deadlines!

For this problem, you should try to formulate as clear of an abstract as you can based on your current progress in project 1. It is OK if you don't have results yet! But hopefully you have a rough idea of what you hope to accomplish and you can articulate why you think the topic is interesting. Every good abstract starts with some context about your problem, then gives some details for your methods, then hints that there could be interesting results. The average length is between 5 and 7 sentences (ish), so this shouldn't take you too long to write!

Hint! If you need some inspiration, take a look at the abstracts of the papers you searched for on ads in homework 2.